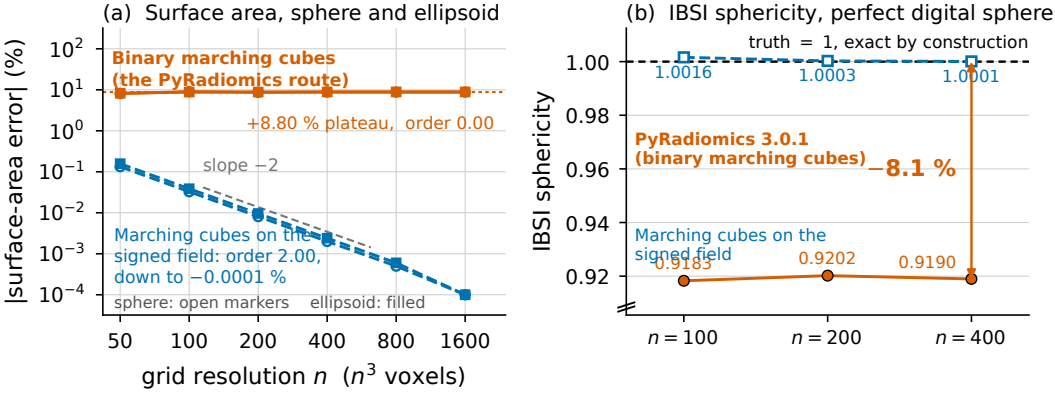
(a) Surface area, sphere and ellipsoid
|surface-area error| (%)
Binary marching cubes
(the PyRadiomics route)
+8.80 % plateau, order 0.00
slope −2
Marching cubes on the
signed field: order 2.00,
down to −0.0001 %
sphere: open markers ellipsoid: filled
grid resolution n (n³ voxels)
(b) IBSI sphericity, perfect digital sphere
IBSI sphericity
truth = 1, exact by construction
1.0016
1.0003
1.0001
PyRadiomics 3.0.1
(binary marching cubes)
−8.1 %
Marching cubes on the
signed field
0.9183
0.9202
0.9190
n = 100
n = 200
n = 400